Rent vs Buy Calculator Specification (India) - Version 2

1. Introduction

Compare financial outcomes of renting vs buying a property over N years in India.

2. Key Assumptions & Defaults

- Gross rental yield (YR): 4.8% p.a. (overrideable per city)
- Stamp duty (SD): 5% of P; Registration fee (RF): 1% of P
- Maintenance cost rate (m_rate): 1% p.a.
- Brokerage on sale (BR): 1% of sale price
- Capital gains tax (CGT): user input (decimal)

3. Inputs

- P: Property cost (■)
- DP: Down payment (■)
- r_loan: Annual loan interest rate (decimal)
- T_loan: Loan tenure (years)
- SD: Stamp duty rate (decimal)
- RF: Registration fee rate (decimal)
- BR: Brokerage on sale rate (decimal)
- g: Expected annual appreciation rate (decimal)
- infl: Annual inflation rate (decimal)
- YR: Gross annual rental yield (decimal)
- re: Rent hike per 11-month period (decimal, e.g., 0.05)
- m_rate: Annual maintenance cost rate (decimal)
- ri: Alternate investment return rate (decimal)
- N: Analysis period (years)

4. Buying Scenario Formulas

4.1 Initial cash outflow: $C\_init = DP + P*(SD + RF)$

4.2 Loan amount: $L = P - DP$

4.3 Monthly EMI: $EMI = [L*(r\_loan/12)*(1+r\_loan/12)^{(12*T\_loan)}]/[(1+r\_loan/12)^{(12*T\_loan)}-1]$

4.4 Outstanding principal after N years: $O\_N = L*(1+r\_loan/12)^{(12*N)} - EMI*(((1+r\_loan/12)^{(12*N)}-1)/(r\_loan/12))$

4.5 Total interest: $Total\_interest = (EMI*12*N) - (L - O\_N)$

4.6 Maintenance each year: $M\_t = P*m\_rate*(1+infl)^{(t-1)}$, sum M_t for $t=1..N$

4.7 Sale proceeds: $P\_sale = P*(1+g)^N$; $NSP = P\_sale*(1-BR) - O\_N$

4.8 Net profit (buy): $Pi\_buy = NSP - C\_init - (Total\_interest + \Sigma M\_t)$

5. Renting Scenario Formulas

5.1 Initial investable capital: $I0 = DP + P*(SD + RF)$

5.2 Annual rent with 11-month hikes: $R\_t = P*YR*(1+re)^{floor(12*(t-1)/11)}$, $t=1..N$

5.3 Monthly rent precision: $r\_m = (P*YR/12)*(1+re)^{floor((m-1)/11)}$, $m=1..12N$

5.4 Future value of capital: $V\_cap = I0*(1+ri)^N$

5.5 FV of annual rents: $FV\_rent\_ann = \Sigma[R\_t*(1+ri)^{(N-t)}]$

5.6 FV of monthly rents: $FV\_rent\_mon = \Sigma[r\_m*(1+ri)^{((12N-m)/12)}]$

5.7 Net profit (rent): $Pi\_rent = V\_cap - FV\_rent$

6. Comparison & Outputs

- Pi_buy, Pi_rent, Difference: $\Delta = Pi\_buy - Pi\_rent$
- Break-even horizon: solve N where $Pi\_buy = Pi\_rent$
- Annualized ROI: $ROI\_buy = (NSP/C\_init)^{(1/N)}-1$; $ROI\_rent = ((V\_cap/FV\_rent)^{(1/N)}-1)$

7. Implementation Notes

- Allow overrides per city/state, input CGT & subsidies
- Visualize cash-flows and net wealth curves
- Include sensitivity analysis for key inputs